

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal

Lecture: CPR 46

Distribution of Body Fluids

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Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0293. List major routes and normal ranges for water intake and loss.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0294. Given the body weight estimate the a) total body water, b) extracellular fluid volume, c) intracellular fluid volume, d) interstitial fluid volume, e) plasma volume, and f) blood volume

SOM.MK.II.BPM1.3.CPR.2.PHYS.0295. Demonstrate the ability to use the indicator dilution principle to measure plasma volume, blood volume, extracellular fluid volume, and total body water, and identify compounds used to measure each volume.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0296. Explain the water movement between intracellular and extracellular fluid compartments caused by increases or decreases in extracellular fluid osmolality.

Recommended Reading

Medical Physiology: Principles for Clinical Medicine with Companion Website Access, R. Rhoades & D.Bell. 6th ed., 2023; Chapter 23

Water Gain and Water Loss

The intake of water is controlled via thirst mechanisms, whereas the removal of water is regulated mainly by the kidneys.

The excretion of water by the kidneys is proportional to the volume of water in the body.

Antidiuretic hormone (ADH) controls the excretion of water by the kidneys.

Water Gain	Water Loss
Liquid consumption 1-2 L	Kidneys 1-2 L
Water in food 0.8-1 L	Skin 0.6 L
Metabolic water 0.3-0.4 L	Lungs 0.3 L
	Fecal loss 0.2 L

Body Fluid Distribution

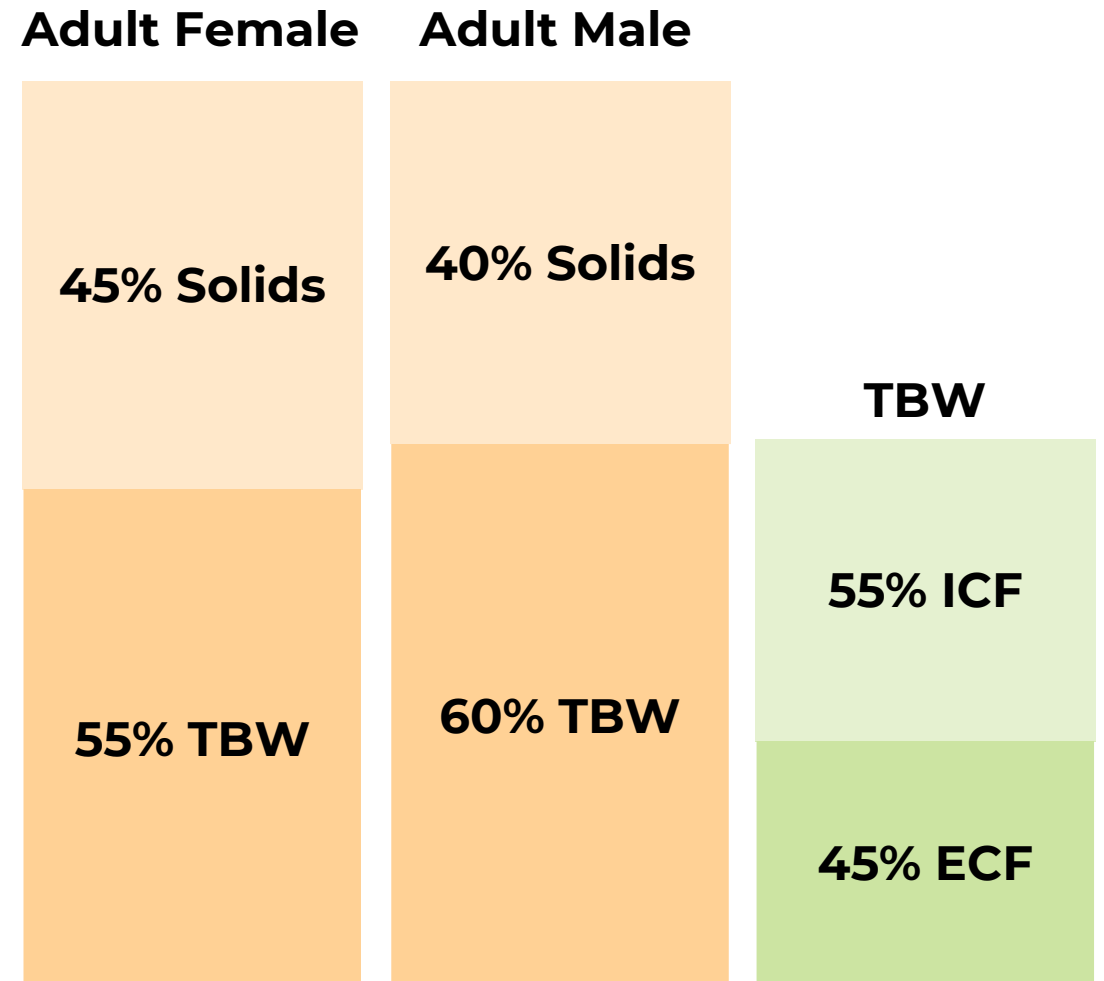
Total body water (TBW) for females is 55% of body weight, and males 60% of body weight.

The intracellular fluid (ICF) makes up 55% of TBW and extracellular fluid (ECF) 45%.

TBW is less in females compared to males, due to the presence of more subcutaneous fat, and less muscle mass

TBW is less in obese individuals.

TBW also decreases with age.



1 kg = 1 L; 1 kg = 2.2 lbs

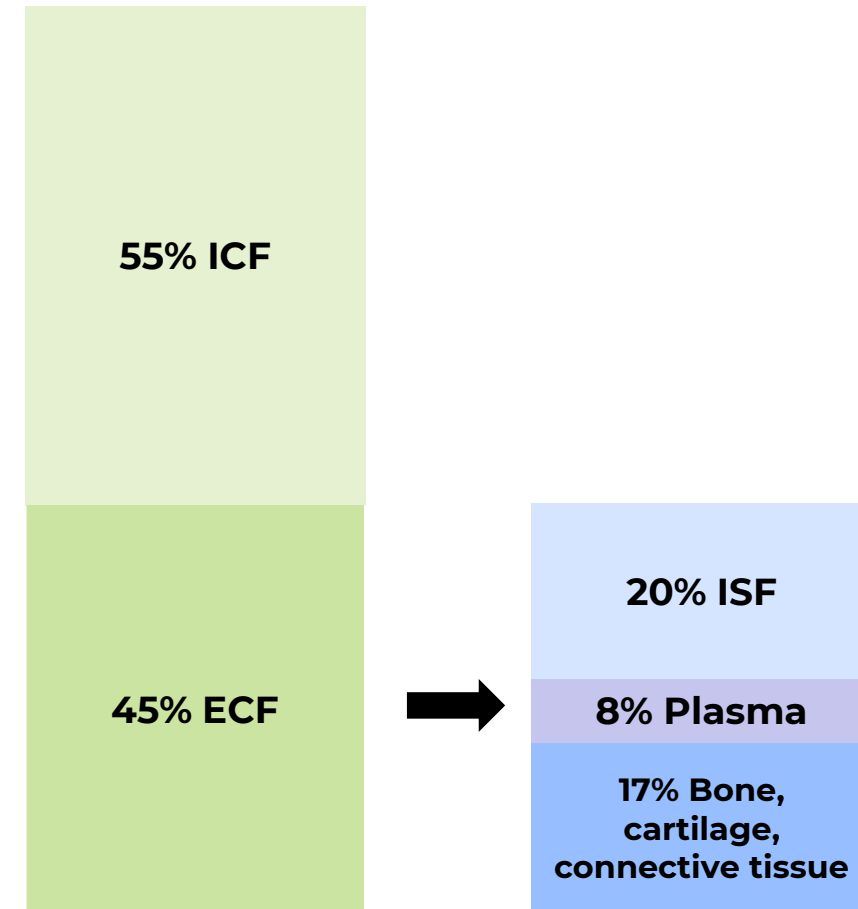
Fluid Spaces of the ECF

The 45% ECF is derived from interstitial fluid (ISF) 20%, plasma 8%, and fluid in bone, cartilage, connective tissue, and transcellular fluid – 17%.

ICF plus ISF and plasma are rapidly interchangeable, while fluid in bone, cartilage, etc, are poorly accessible.

As a result, swift changes in the distribution of water occur between the ICF and rapidly interchangeable ECF (plasma and ISF), making up 83% of TBW (55 + 20 + 8).

Therefore, $ICF = 2/3^{rds}$ TBW; *rapid* $ECF = 1/3^{rd}$ TBW



Example 1

In a 70 kg man with a hematocrit of 45%, determine the TBW, ICF, ECF, ISF, plasma volume(PV), and blood volume(BV)

$$\text{TBW} = 60\% \text{ of body weight} = 0.6 \times 70 = 42 \text{ L}$$

$$\text{ICF} = 55\% \text{ of TBW} = 0.55 \times 42 = 23.1 \text{ L}$$

$$\text{ECF} = 45\% \text{ of TBW} = 0.45 \times 42 = 18.9 \text{ L}$$

$$\text{ISF} = 20\% \text{ of TBW} = 0.20 \times 42 = 8.4 \text{ L}$$

$$\text{PV} = 8\% \text{ of TBW} = 0.08 \times 42 = 3.4 \text{ L}$$

$$\text{BV} = 3.4 / 1 - 0.45 = 3.4 / 0.55 = 6.2 \text{ L}$$

$$\text{BV} = \frac{\text{PV}}{(1 - \text{Hct})}$$

$$\begin{aligned} \text{Also, ICF} &= \text{TBW} - \text{ECF} \\ \text{ECF} &= \text{TBW} - \text{ICF} \end{aligned}$$

Fluid Compartment Volume Measurement

An indicator-dilution method is utilized.

In this case, indicators can be used to estimate the volumes of various compartments.

TBW

- Isotopic water (heavy water, D₂O)
- Tritiated water (HTO)
- Antipyrine

ECF

- Radioactive sodium
- Sulfate
- Mannitol
- Inulin

Plasma

- Radioactive albumin
- Evans blue

For ICF and ISF, estimation is done indirectly:

$$\text{ICF} = \text{TBW} - \text{ECF}$$

$$\text{ISF} = \text{ECF} - \text{PV}$$

$$\text{BV} = \frac{\text{PV}}{(1 - \text{Hct})}$$

Compartment Volume Formula

After administration of a known amount of the indicator (dye) and allowing adequate time for equilibration, a plasma sample is collected, and the plasma concentration is determined.

The compartment volume distribution can then be calculated using the formula:

$$V = \frac{Q}{C}$$

V = Compartment volume

Q = Amount injected – amount excreted

C = Concentration of the indicator in plasma after equilibration

Example 2

A 60 kg man is administered 100 mL of D2O, of which 10% is excreted. After equilibration, the plasma concentration is 2.5 mL/L.

1. What body fluid compartment volume can be calculated?
2. Calculate the compartment volume.

Answer:

1. TBW

$$\begin{aligned} 2. V &= Q / C \\ &= 100 - 10 / 2.5 \\ &= 90 \text{ mL} / 2.5 \text{ mL/L} \\ &= 36 \text{ L} \end{aligned}$$

*A TBW of 36L in a 60 kg man
implies that TBW is 60% of his
body weight ($36 / 60 \times 100 = 60\%$)*

Example 3

Following the administration of 5 mCi of (radioactive) ^{131}I -albumin to a patient, 0.5% is metabolized and excreted. After equilibration, plasma concentration of the marker is 2 mCi.

1. Which fluid compartment can be measured?
2. Calculate the compartment volume.

Answer:

1. Plasma volume
2.
$$\begin{aligned} V &= Q / C \\ &= 5 - 0.025 / 2 \text{ (0.5\% of 5 is 0.025)} \\ &= 4.475 / 2 \\ &= 2.5 \text{ L} \end{aligned}$$

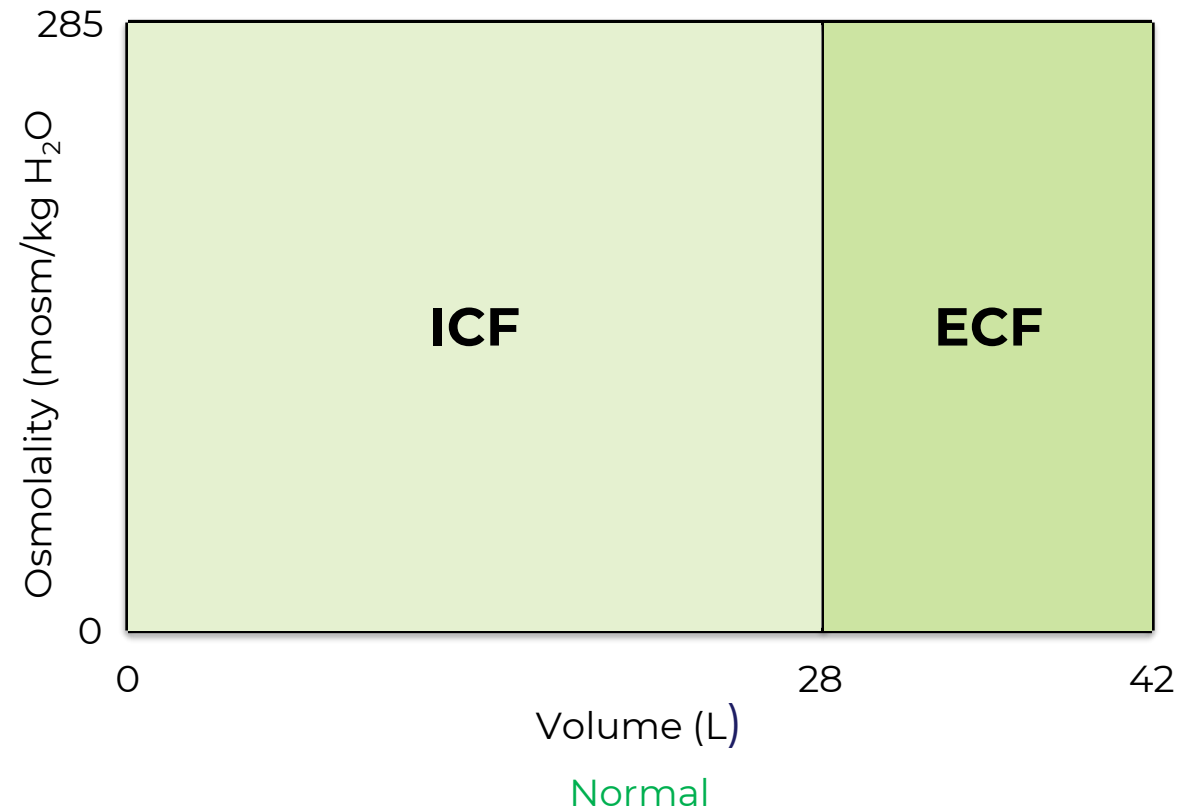
The Darrow-Yannet Diagram

The Darrow-Yannet diagram is a schematic that allows for the visualization of volume and osmolality changes in the ICF and ECF.

Volume changes affect the ECF volume first.

ICF volume changes only when there is a change in ECF osmolality.

The osmolality of the ICF and ECF are equal due to osmotic equilibration.



Volume and Osmolality Changes in Fluid Compartments

Volume changes take the form of expansion or contraction (reduction) in fluid compartments.

These changes can be isosmotic, hyperosmotic, or hyposmotic.

	Volume Expansion	Volume Contraction
Isosmotic	Isosmotic volume expansion	Isosmotic volume contraction
Hyperosmotic	Hyperosmotic volume expansion	Hyperosmotic volume contraction
Hyposmotic	Hyposmotic volume expansion	Hyposmotic volume contraction

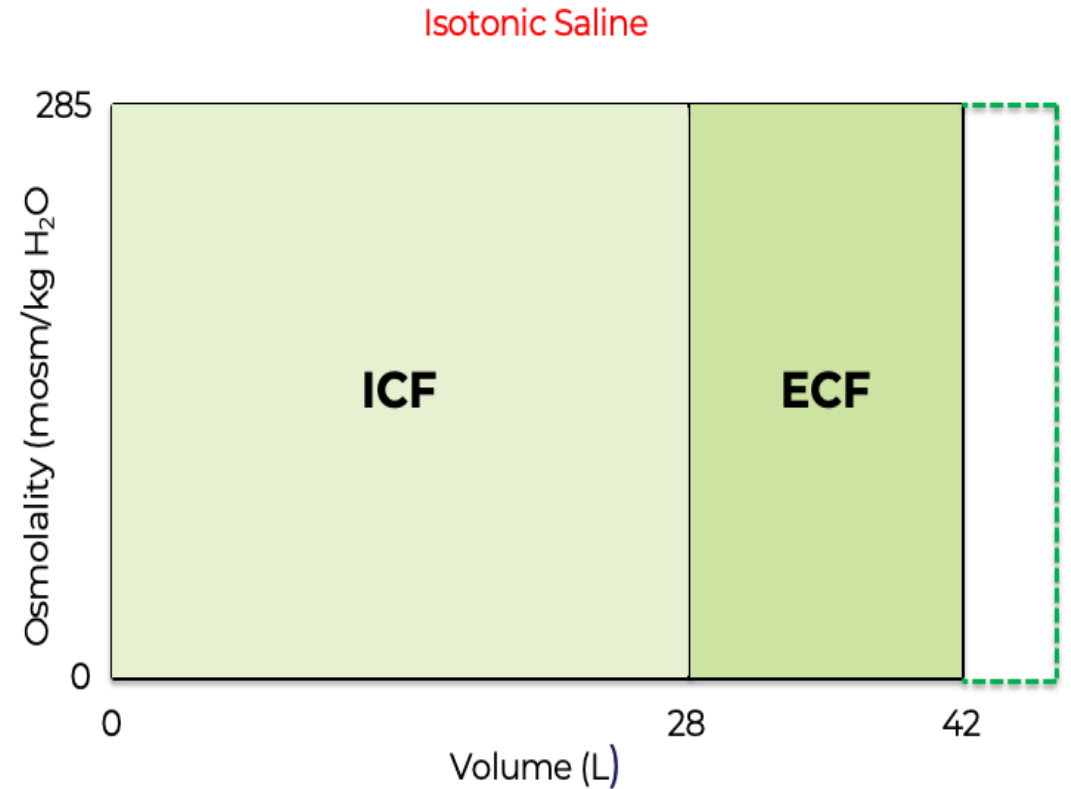
$$\text{Plasma osmolality} = 2 \times \text{Na (mEq/L)} + \text{glucose (mg/dL)} / 18 + \text{BUN (mg/dL)} / 2.8$$

Isosmotic Volume Expansion

Expansion of ECF volume by the addition of isotonic saline to it.

Example: The administration of **0.9% NaCl solution (isotonic saline)** to the ECF results in the expansion of ECF volume only.

The osmolality does not change since this solution has the same osmolality as normal body fluids.

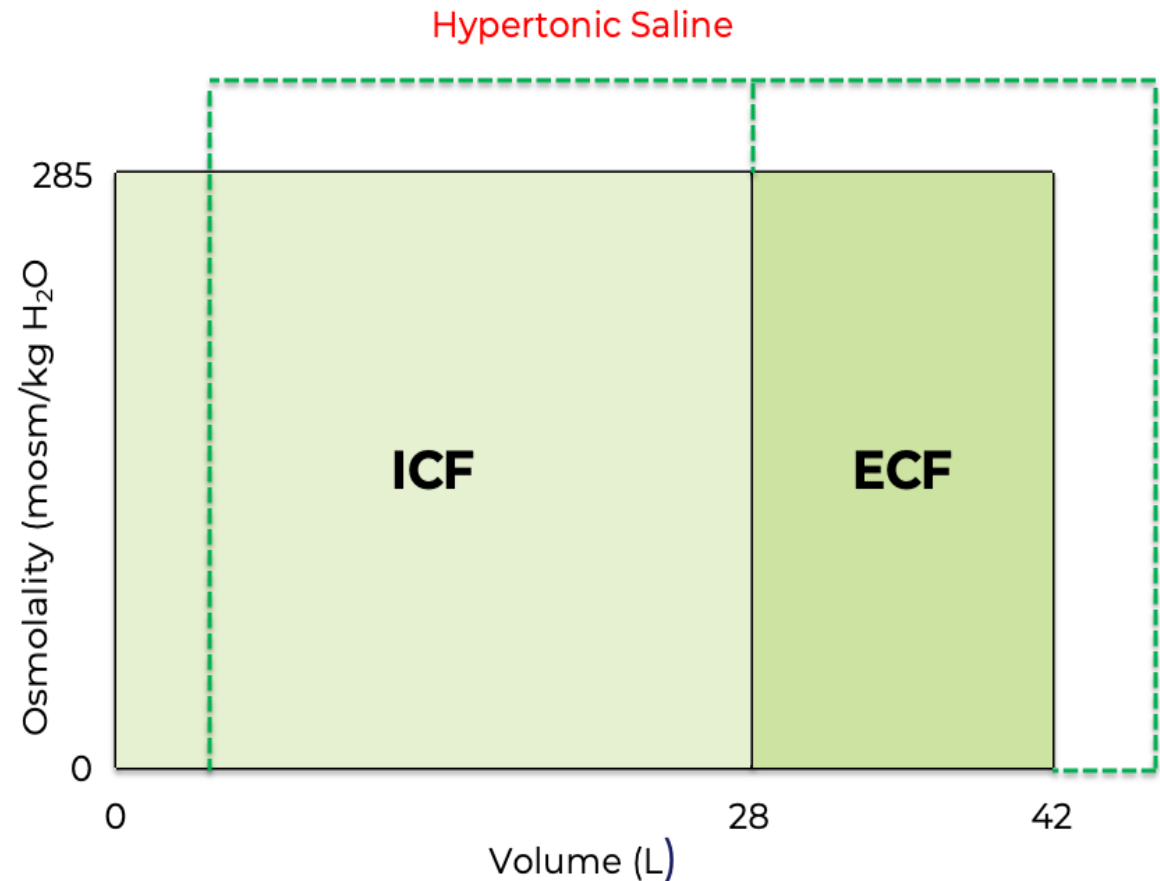


Hyperosmotic Volume Expansion

Expansion of ECF volume by the addition of hypertonic saline to it.

Example: **High salt intake or hypertonic saline (3% NaCl)** leads to an increase in ECF osmolality.

Water moves from the ICF to the ECF, resulting in the expansion of ECF volume.

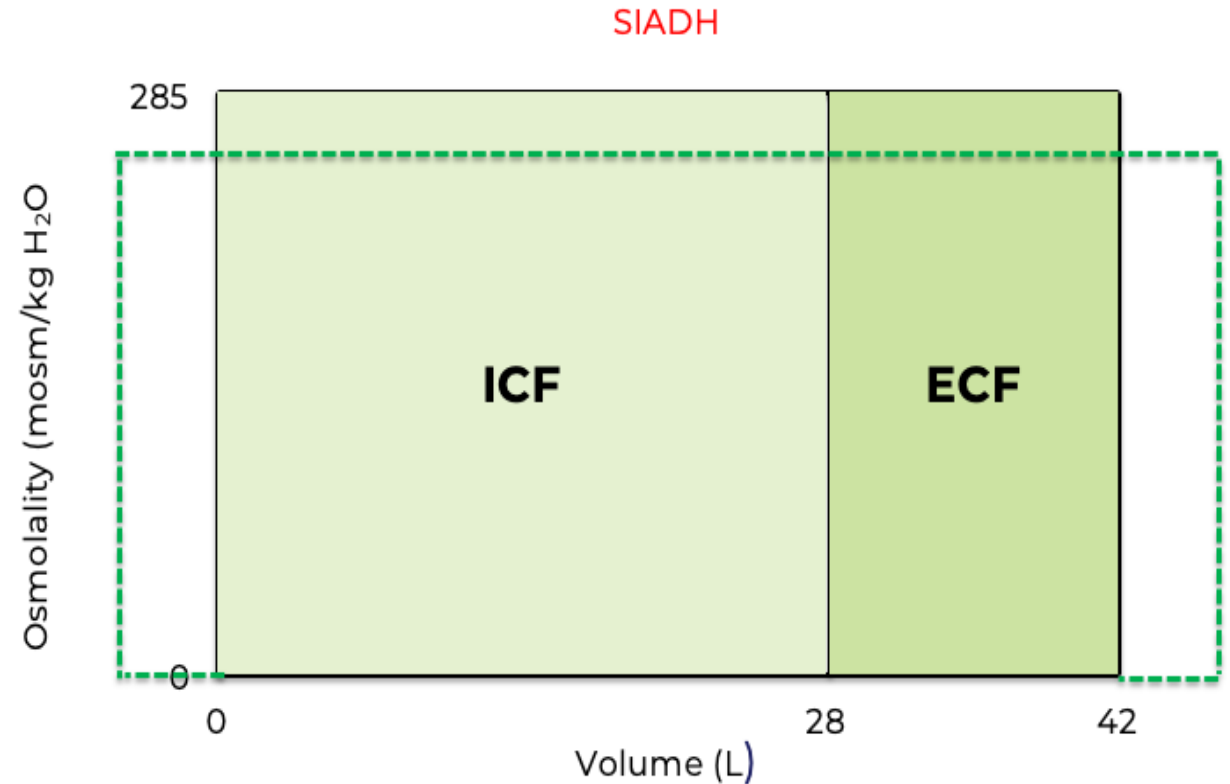


Hypotonic Volume Expansion

Expansion of ECF volume by the addition of hypotonic fluid to it.

Example: **Ingestion of large amounts of water or SIADH** (increased reabsorption of water) increases the volume of water in the ECF while decreasing osmolality.

Water movement into the ICF also increases ICF volume and decreases osmolality.

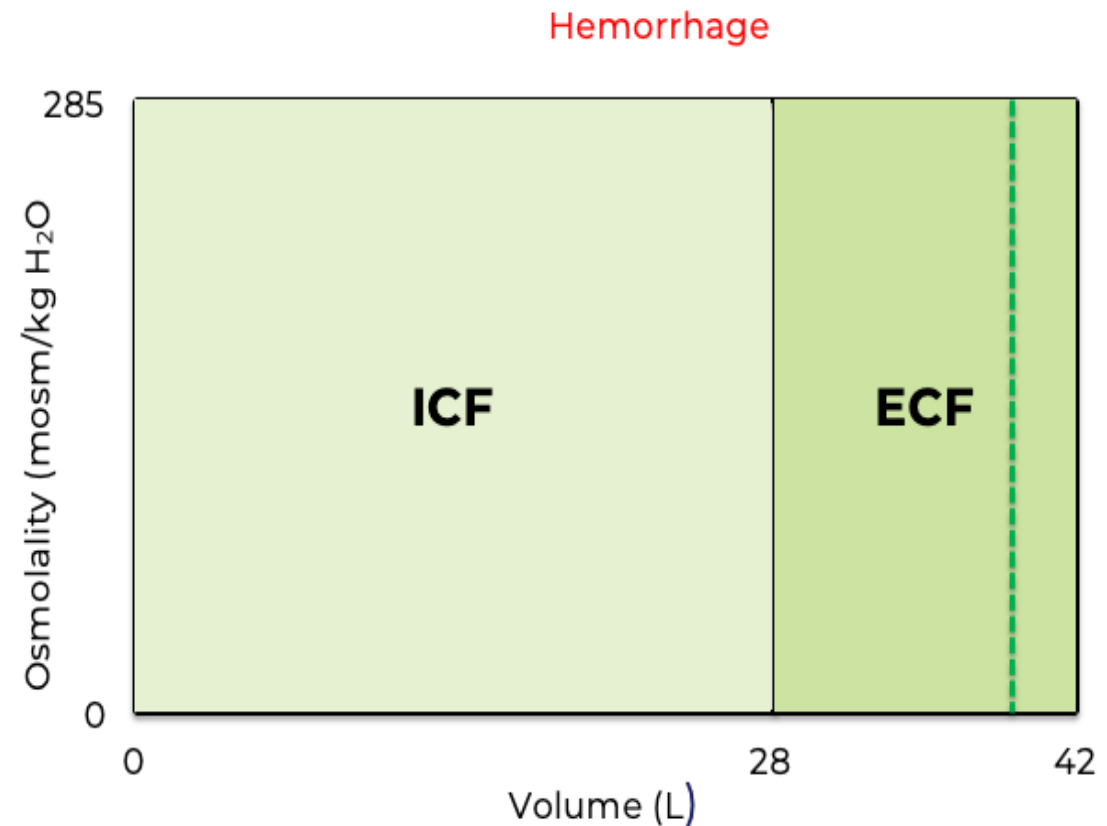


Isosmotic Volume Contraction

Reduction in ECF volume by loss of isosmotic fluid from it.

Acute blood loss, vomiting, and diarrhea (certain types) are examples of conditions that lead to reduced fluid volume.

The osmolality stays the same because isotonic fluid is lost.

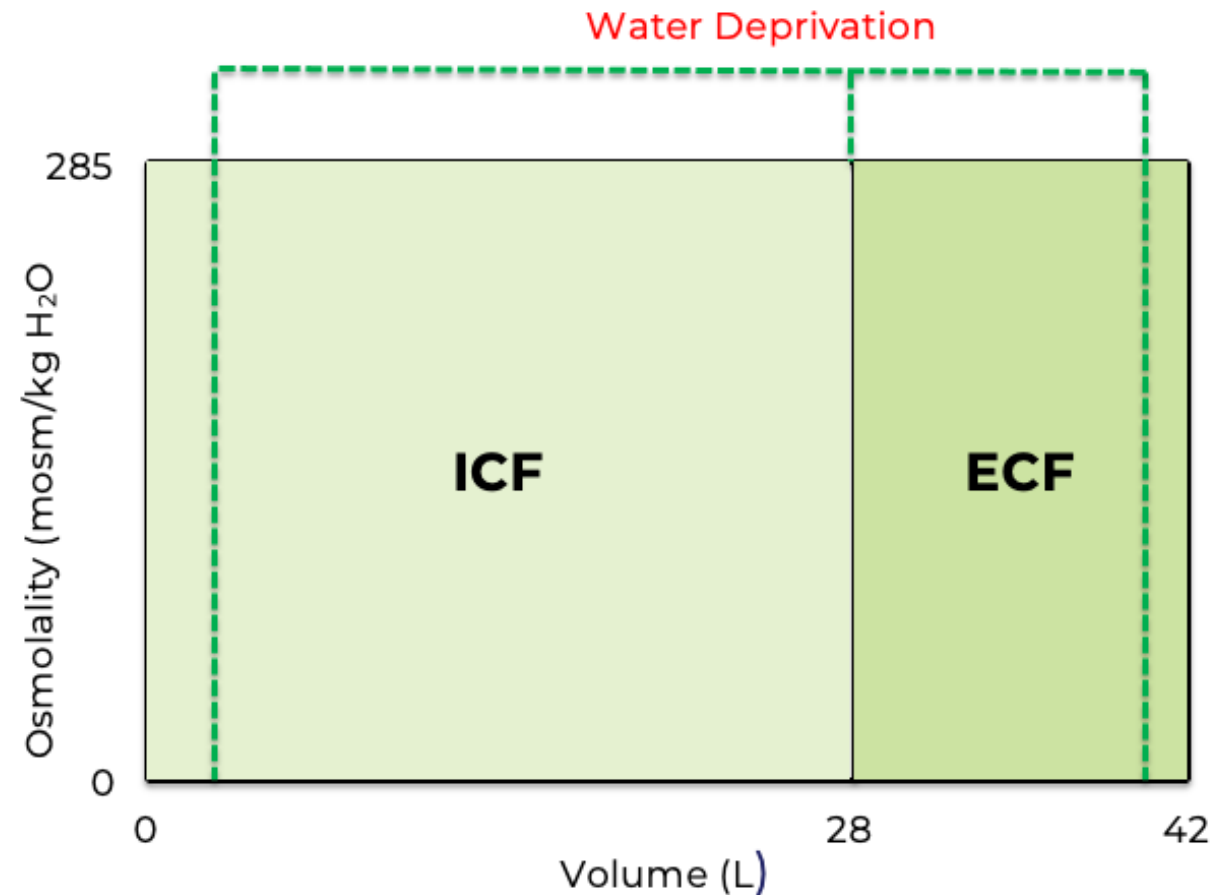


Hyperosmotic Volume Contraction

Reduction in ECF volume by loss of hypotonic fluid from it.

Water deprivation and severe sweating are examples of conditions that can cause hyperosmotic volume contraction.

The loss of hypotonic fluids leads to an increase in the ECF osmolality. This results in the movement of water from the ICF to the ECF to equilibrate both compartments.

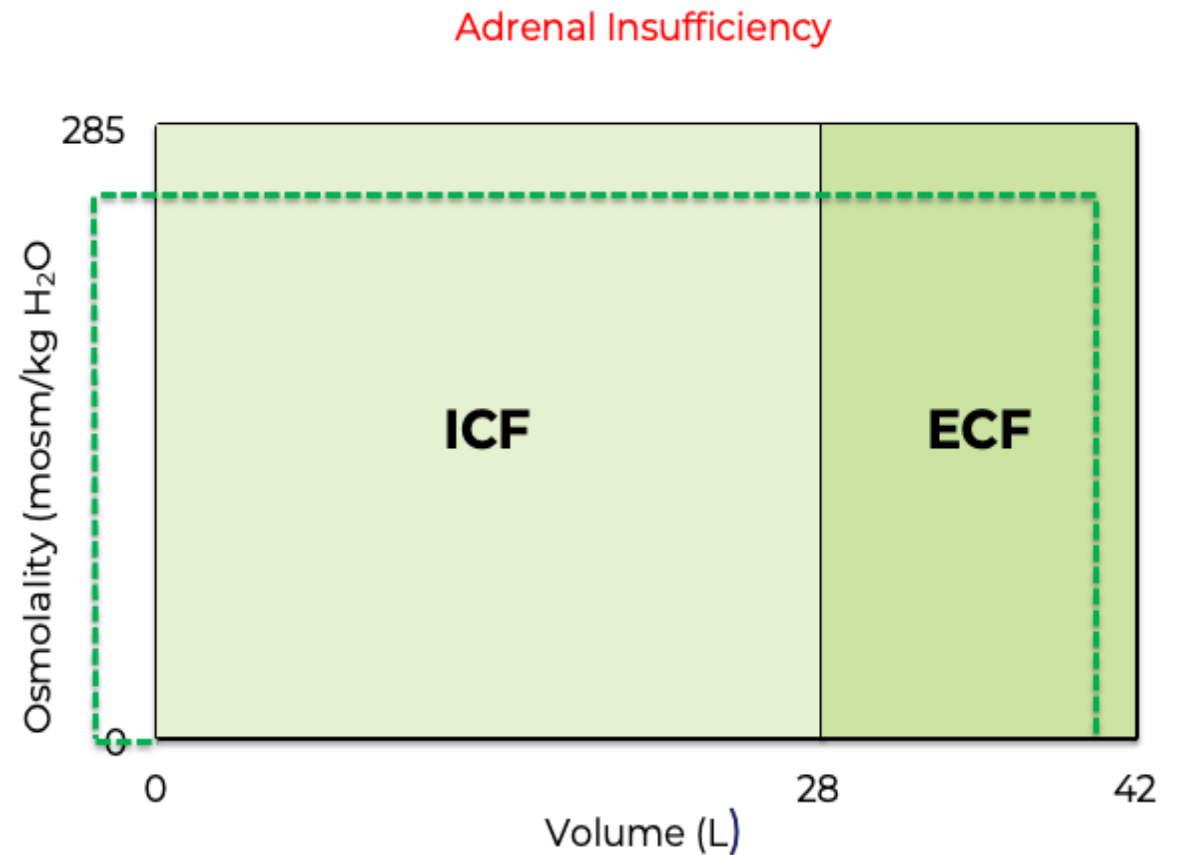


Hypotonic Volume Contraction

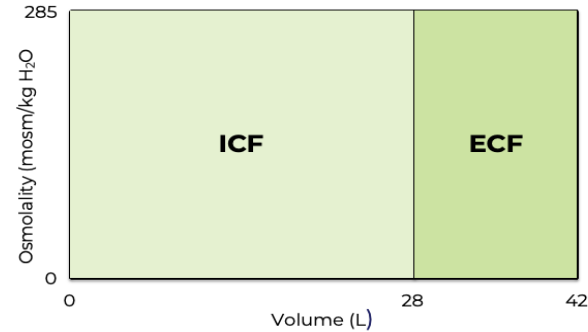
Reduction in ECF volume with a decrease in sodium concentration.

The **lack of aldosterone in adrenal insufficiency** can lead to hypotonic volume contraction.

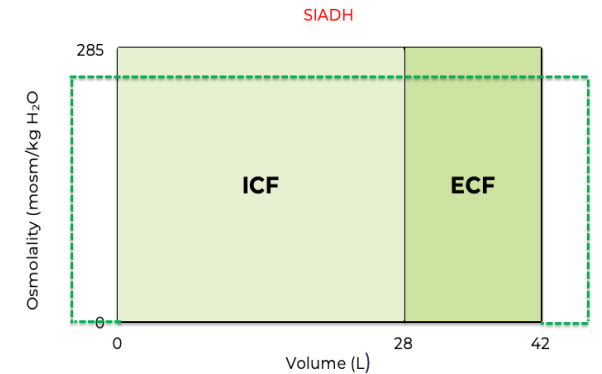
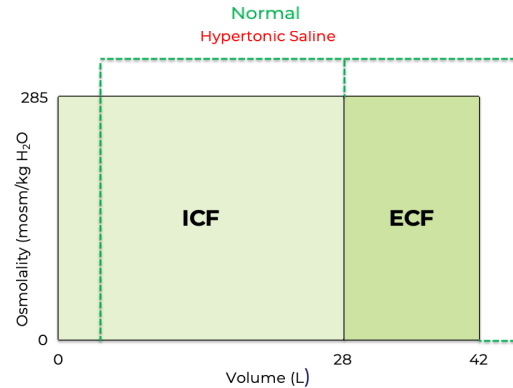
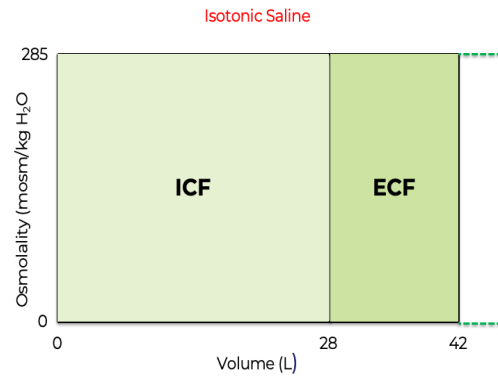
The reduced reabsorption of sodium leads to a decrease in ECF osmolality. Water then moves from the ECF to the ICF.



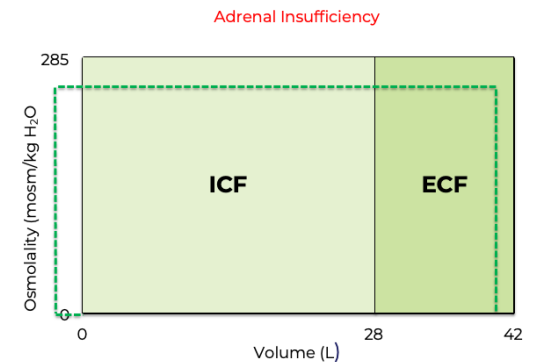
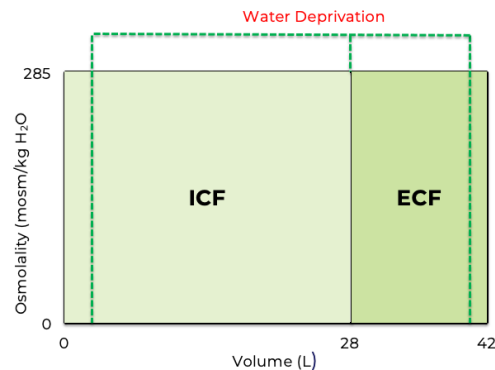
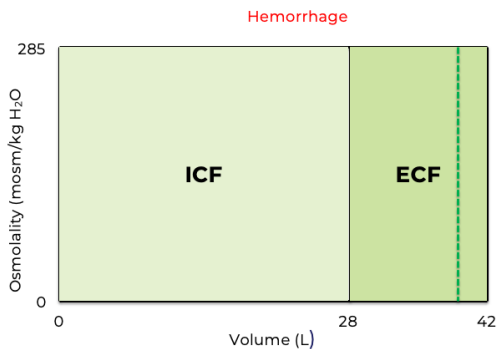
Summary - Volume and Osmolality Changes



Volume Expansion



Volume Contraction



Intravenous Fluids

Isotonic

0.9% NaCl, Ringers solution

Hypotonic

0.45% NaCl, 5% dextrose

Hypertonic

5% dextrose in 0.9% NaCl, 3% NaCl